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COLLEGE OF INFORMATICS AND COMPUTING INNOVATIONS (CICI)
Tiwi Campus, Barotac Nuevo, Iloilo | Capstone Project Research Group

SECUREFACULTY: STANDALONE ONE-CLICK SETUP

Zero-Docker & Zero-XAMPP Native Windows Deployment Guide

DOCUMENT: Operations Guide	ENVIRONMENT: Native Windows 10/11	DATABASE: Embedded SQLite Engine

Summary: This official guide allows you to run the complete SecureFaculty system directly on any client Windows computer without Docker, without WSL2, and without installing database servers like MySQL or XAMPP. Utilizing native Windows PHP and the zero-configuration embedded SQLite engine, the portal boots instantly in **under 3 seconds**.

1. System Prerequisites (Pre-installed on Client PC)

The host computer only requires standard native PHP and Python runtimes:

Runtime Component	Required Version	Verification Command
PHP for Windows	PHP 8.0, 8.1, 8.2, or 8.3	php -v
Python for Windows	Python 3.10, 3.11, 3.12, or 3.13	python --version

2. Install Required Python Libraries

Open Command Prompt or PowerShell in the project folder and run:

```
pip install -r requirements.txt
# Installs: pillow (images), pypdf (PDF parsing), python-docx, requests, reportlab (PDF engine)
```

3. Configure Embedded SQLite (Single-Line Change)

Configure the database driver to use self-initializing SQLite instead of MySQL:

Step A Open **app/config.php** in any text editor (Notepad, VS Code).

Step B Locate **Line 21** and set the database driver to 'sqlite':

```
// Line 21 in app/config.php
define('DB_DRIVER', 'sqlite');
```

** Note: SQLite requires zero database server setup. The database file **storage/securefaculty.sqlite** and all tables/schemas are created automatically on the very first HTTP request.*

4. Launch the System (One-Click)

Start the web portal using either method:

Method A (One-Click)	Double-click **start_securefaculty.bat** in the root project folder.
Method B (Command Line)	Run in terminal: python run_server.py

5. Open Portal in Web Browser

Open Google Chrome or Microsoft Edge and navigate to:

Live Web URL: http://localhost:8000/ (or http://127.0.0.1:8000/)

SECUREFACULTY: EVALUATION PROTOCOLS & TROUBLESHOOTING

Account Credentials, Verification Checklist, and System Diagnostics

6. Pre-Configured Demonstration Accounts

Log in using the pre-seeded credentials with role-based access control (RBAC):

Role / Clearance	Username	Password	Permissions & Scoped Features
Dean (Admin)	**dean**	**admin123**	Administrative clearance, centralized dashboard, user registry, compliance audit ledger, department-wide vault
Faculty Member	**mikhail**	**faculty123**	Personal encrypted vault, co-authored publications, certificate ingestion pipeline
Faculty Member	**raejan**	**faculty123**	Personal encrypted vault, document ingestion gateway, co-authorship badges
Faculty Member	**eljay**	**faculty123**	Personal encrypted vault, publication archive, multi-author research records

7. Demonstration & Feature Verification Checklist

Perform these tests during evaluation to demonstrate key system capabilities:

Feature	Navigation Path	Expected Behavior & Compliance Standard
Smart Pipeline	Click **Ingestion Pipeline** in nav	Click 'Load Demo Sample' -> Stage 1 to 4 telemetry displays OCR metadata extraction & H-I-T-L validation.
AES-256 Vault	Click **Digital Vault**	Zero plaintext storage at rest. Clicking 'Decrypt & View' recovers binary payload in volatile RAM.
Co-Authorship	Search for 'SECUREFACULTY'	Attributed to Mikhail, Rae Jan, and Eljay simultaneously with interactive co-author distinction badges.
Audit Ledger	Log in as Dean -> **Audit Trail**	Immutable compliance ledger tracking IP, timestamp, role, and actions in full compliance with RA 10173.
Portrait A4 PDF	Click **Accreditation Report** -> PDF	Downloads official standalone A4 Portrait PDF with university letterhead, borders, and running footers.

8. Troubleshooting & Windows Diagnostics

- **'php' or 'python' is not recognized:** Re-run the installer and check *'Add to PATH'*, or restart Command Prompt.
- **Port 8000 already in use:** Launch on a different port: php -S 127.0.0.1:8080 router.php and browse to http://localhost:8080/.
- **SQLite PDO extension missing:** In php.ini, ensure extension=pdo_sqlite and extension=sqlite3 are uncommented.
- **OpenSSL extension missing:** In php.ini, ensure extension=openssl is uncommented for AES-256 vault encryption.
- **ReportLab PDF compilation:** Ensure ReportLab is installed: python -c "import reportlab; print('ReportLab OK')".

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